Software and Web Development (SWD 101)

Lesson 1: Introduction to Software Development

LESSON OBJECTIVES

By the end of this lesson, students should be able to:

- Define software and software development.
- Explain the software development life cycle (SDLC).
- Identify different types of software.
- Describe the roles of software developers.
- Explain the importance of software engineering.
- Recognize common programming languages and development tools.

1.1 What is Software?

Software is a collection of programs, instructions, and data that tells a computer how to perform specific tasks. Unlike hardware, software is intangible and exists in digital form.

Examples of software include:

- Microsoft Word
- Google Chrome
- WhatsApp
- Adobe Photoshop
- Spotify
- Banking applications
- School management systems
- E-learning platforms

1.2 Types of Software

Software is generally classified into three primary categories:

1. System Software

System software manages the computer's hardware and provides a platform for other software to run.

Examples: Windows, Linux, macOS, Android

Key Functions:

- Manages computer memory
- Controls hardware peripherals and devices
- Runs background applications
- Handles file storage and system security management

2. Application Software

Application software helps end-users perform specific tasks, productivity work, or entertainment.

Examples: Microsoft Excel, Google Docs, Zoom, Web browsers, Media players, Learning Management Systems (LMS)

3. Programming Software

Programming software provides tools to help developers create, debug, and maintain other software and applications.

Examples: Visual Studio Code, PyCharm, Eclipse, Android Studio

1.3 What is Software Development?

Software development is the end-to-end process of designing, creating, testing, deploying, and maintaining software applications to solve real-world problems.

Software development involves: Planning, Designing, Coding, Testing, Deployment, and Maintenance.

1.4 Importance of Software Development

Software development has transformed key global sectors:

- **Education:** Online learning, student portals, and virtual classrooms.
- **Healthcare:** Electronic medical records, hospital management systems, and telemedicine.
- **Banking:** Mobile banking, ATM software, and secure online payment gateways.
- **Business:** Inventory systems, automated payroll, and e-commerce platforms.
- **Government:** Digital identity systems, tax management, and e-government portals.

1.5 Software Development Life Cycle (SDLC)

The SDLC is a structured process used by software engineering teams to build high-quality software efficiently.

PHASE	DESCRIPTION & KEY ACTIVITIES
Phase 1: Planning	Identify the core problem, project objectives, resource allocation, and project scope.
Phase 2: Requirements Analysis	Gather and document complete technical and operational user requirements.
Phase 3: System Design	Architect software systems, user interface (UI) mockups, database schemas, and workflow logic.
Phase 4: Implementation	Write production source code using chosen programming languages, libraries, and frameworks.
Phase 5: Testing	Systematically verify performance, validate requirements, and eliminate software bugs/defects.
Phase 6: Deployment	Release the finalized software application to production servers or app stores for users.
Phase 7: Maintenance	Perform post-release updates, security patching, feature updates, and performance optimization.

1.6 Roles in a Software Development Team

Modern software development requires specialized team roles collaborating together:

- **Project Manager:** Oversees overall project execution, planning, timelines, and team resources.
- **Business Analyst:** Bridges stakeholder expectations by gathering and documenting business requirements.
- **UI/UX Designer:** Crafts intuitive user experiences (UX) and visual user interfaces (UI).
- **Front-End Developer:** Builds client-facing application interfaces and user interactions.
- **Back-End Developer:** Implements core server logic, data management, and business APIs.
- **Database Administrator (DBA):** Architects, secures, and optimizes database infrastructure.
- **Quality Assurance (QA) Engineer:** Executes systematic test plans to ensure high software quality and stability.
- **DevOps Engineer:** Manages CI/CD pipelines, automated deployments, and cloud infrastructure.

1.7 Common Programming Languages

Select programming languages based on target platform and specialized application domains:

LANGUAGE	PRIMARY COMMON USES
Python	Web development, artificial intelligence, machine learning, automation, and data science.
JavaScript	Interactive web front-ends, modern web applications, and full-stack development (Node.js).
Java	Enterprise backend systems, cross-platform applications, and native Android app development.
C++	Operating systems, game engines, high-performance applications, and embedded systems.
C#	Enterprise desktop applications, web platforms (.NET), and game development (Unity).
PHP	Dynamic server-side web application backends and content management platforms.
Kotlin	Modern native Android application engineering.
Swift	Native iOS, macOS, and Apple platform app development.

1.8 Development Tools

Essential developer toolkits include:

- **Code Editors & IDEs:** Visual Studio Code, PyCharm, Eclipse, Android Studio
- **Version Control & Collaboration:** Git, GitHub, GitLab
- **Runtimes & Frameworks:** Python, Node.js, Django, React, Express
- **Databases:** PostgreSQL, MySQL, SQLite, MongoDB
- **DevOps & Testing Tools:** Docker, Postman, Jest

1.9 Key Terms Glossary

Software: Collection of programs, procedures, and data that perform operational tasks on a computer.

Hardware: The physical microelectronics, mechanical, and tangible components of a computer system.

Algorithm: A finite, well-defined sequence of step-by-step instructions to solve a specific problem.

Program: Executable instructions written in code designed to perform computer operations.

Compiler: Software that converts human-readable source code directly into machine code.

Framework: A foundational structure providing generic, reusable functionality for building software applications.

✍️ PRACTICAL EXERCISE

1. List five examples of system software.
2. List five examples of application software.
3. Identify and describe the seven phases of the SDLC.
4. Explain the conceptual and physical differences between hardware and software.
5. Research three software applications used in modern education and summarize their primary purpose.

📝 HOMEWORK ASSIGNMENT

Topic: Impact of Software in Modern Education

Develop a comprehensive one-page academic report (500–700 words) describing how software development has transformed educational delivery in your country. Include at least three concrete, real-world case studies or software examples.

🧪 KNOWLEDGE CHECK QUIZ

1. What is software?
2. Name the three major categories of software.
3. What does the acronym **SDLC** stand for?
4. Which SDLC phase focuses on identifying and documenting user needs?
5. Name two programming languages heavily utilized for web development.
6. What primary responsibilities fall under the role of a QA Engineer?
7. Give two distinct examples of system software.
8. Why is post-deployment maintenance essential in the software lifecycle?
9. Distinguish between application software and programming software.
10. List three fundamental development tools used daily by software engineers.

➡️ **Coming Up Next — Lesson 2: Introduction to Web Development**

In our next lecture, we will explore how the global Web works, examining client-server architecture, browser rendering, web servers, HTTP/HTTPS protocols, domain name systems (DNS), web hosting, and the technical breakdown of front-end vs. back-end stacks.